

Generalizing Personalized Federated Graph Augmentation via Min-max Adversarial Learning

Liang Zhang*
zhangliang@sribd.cn
Shenzhen Research Institute of Big
Data
Shenzhen, Guangdong, China

Tao Long*
longtao2021@email.szu.edu.cn
College of Computer Science and
Software Engineering, Shenzhen
University[‡]
Shenzhen, Guangdong, China

Yang Liu
yang.liu2@dukekunshan.edu.cn
Data Science Research Center,
Dukekunshan University
Kunshan, Jiangsu, China

Lei Zhang[†]
leizhang@szu.edu.cn
College of Computer Science and
Software Engineering, Shenzhen
University
Shenzhen, Guangdong, China

Laizhong Cui
cuilz@szu.edu.cn
College of Computer Science and
Software Engineering, Shenzhen
University
Shenzhen, Guangdong, China

Qingjiang Shi
shiqj@tongji.edu.cn
School of Software Engineering,
Tongji University
Yangpu, Shanghai, China

Abstract

Federated learning (FL) enables the training of a global machine learning model among multiple local clients in a collaborative fashion without directly sharing the details of their data. Due to this advantage, it has been utilized in a wide range of applications where privacy is a critical concern and has attracted great attention for graph representation learning (GRL). Despite the offered advances, there still exist two major challenges in the FL for GRL across distributed graph data, including heterogeneity and complementarity. In order to tackle these challenges, a novel personalized federated graph augmentation (PFGA) framework is proposed in this work. Unlike existing techniques, it utilizes generative models as bridges to enable information sharing among clients, thereby facilitating the collaborative training of GRL models. Instead of directly using the generative model trained on each client individually, we aggregate them into the globally generative model to gain a global view of the entire graph, which effectively alleviates the heterogeneity and complementarity issues simultaneously. We formulate the training of the generative and GRL models as a min-max adversarial learning problem and theoretically prove the convergence. Furthermore, the effectiveness of the method is demonstrated using experimental results on six real-world datasets.

CCS Concepts

• **Computing methodologies** → **Distributed artificial intelligence; Learning latent representations.**

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Graph Augmentation; Federated Learning; Min-max Learning.

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1 Introduction

Graph representation learning (GRL) algorithms like the graph neural networks have gained impressive performance in learning low-dimensional representations of nodes in non-Euclidean graph data, which are further utilized for downstream tasks, such as node classification [32], link prediction [2], and graph classification [37]. While the GRL methods for centralized learning scenarios, where all graph data are stored and trained in a centralized fashion, have already been sufficiently studied in existing literature, there remains a broad range of industrial scenarios in which the graph data are distributed across devices in different organizations such as financial transaction networks [11] and recommender systems [36]. Therefore, a GRL method that can capture representative information collaboratively from the isolated data islands is urgently needed. However, it is a challenging task to exploit distributed graphs to train a global GRL model without leakage of private information.

Recently, federated learning (FL) has emerged as a powerful tool to facilitate collaborative training based on distributed data sources [7, 33], which preserves data privacy through aggregating the updates from each client to the global model on the public server. Despite the advances, the FL technique still suffers from the *heterogeneity* difficulty induced by the Non-Independent Identically

*The two authors contribute equally to this work.

[†]Corresponding author.

[‡]Primary completion affiliation.

Distribution (non-IID) of data from different clients [12, 19]. It can jeopardize the stability of model convergence and consequently impact the quality in practical applications [7]. In existing literature, this problem is tackled by several approaches, such as incorporating regularization terms to avoid over-fitting to local data [15] and learning a personalized mask for each client to select only useful parameters from the global model [1]. However, this challenge is not sufficiently addressed in extremely heterogeneous scenarios, where the quality of the collaboratively trained model can even be inferior to the models trained locally for each client.

When FL is utilized in GRL tasks, a specific challenge is *complementarity*, which is induced by the potential correlations between graph data distributed in different clients. Specifically, there might exist overlapped nodes across the graphs in different clients while for each client the information of several edges is missing. As a consequence, the graph data on each client lacks important neighborhood information, which can significantly compromise the training of the GRL model in each client. Some recent works [29, 39] attempt to augment the missing nodes by requesting the node information. For instance, FedSage+ [39] trained a neighbor generator in each client that attempted to generate missing cross-client edges via requiring the embedding of nodes from other clients so as to finding out the closest ones. However, these methods may violate data privacy constraints and induce heavy communication overhead across clients.

To address the aforementioned challenges, we propose a novel personalized federated graph augmentation framework (PFGA) consisting of GRL models and auxiliary generative models. The generative models are trained to generate synthetic graph samples with minimum sufficient information to enhance the performance of the GRL models. In contrast to existing works that directly aggregate the GRL models trained on each client, the proposed PFGA framework leverages the generative models to aggregate the local-client information. To address the heterogeneity problem, the locally generative models are aggregated to form the globally generative model while the GRL models are always kept personalized to fit the characteristics of local graph data. This design also features better data privacy as we only share the auxiliary generative models, while keeping data and key GRL models confidential. As the globally generative model has a global view over the data distribution in all clients, the augmented samples have the potential to capture missing information due to missing edges across different clients. To alleviate the complementarity problem, instead of augmenting missing edges directly, we attempt to remedy the missing information by augmenting new graphs from globally generative models. We also notice that the locally generative model in each client can provide augmented graph samples with an independent view of the global graph. Therefore, we also propose the clustered version of the PFGA framework, which assigns locally generative models into clusters and aggregates them into clustered globally generative models. This provides an option that preserves more detailed information to fit the utilization in specific scenarios.

We formulate the training of GRL and generative models as a min-max adversarial learning problem and provide proofs such that its convergence under a federated learning context can be guaranteed theoretically. Although two recent works [22, 31] have also provided the convergence analysis of min-max optimization,

they share all parameters into the server and assume that the updating of min-max parameters is synchronous, i.e., the parameters of min problem and max problem are updated in the server synchronously. In our PFGA framework, we update the max problem in each local client and update the min problem in the server. The updating of min and max problems is asynchronous which brings a different theoretical analysis framework. We have generalized the convergence from convex-concave objectives to non-convex-non-concave objectives in the min-max optimization. This is enabled by decoupling the optimization of GRL models from that of globally generative models, such that PFGA updates the globally generative model using stochastic gradient descent similar to standard FL algorithms while optimizing the personalized GRL models in parallel with low communication cost. For the non-convex-non-concave objective, we analyze the accumulated gradients with the alternative stochastic gradient descent algorithm, which reduces the initial errors linearly and is upper bound by the variance of stochastic gradients. Our main contributions can be summarized as follows.

- We proposed a novel PFGA framework, which generates augmented graph samples using the globally generative model to enable information sharing between clients and facilitate the effective personalized training of GRL models.
- We formulated the training of GRL and generative models as a min-max adversarial learning problem and provided theoretical proofs of its convergence under an asynchronous federated learning context.
- We conducted extensive experiments to verify the effectiveness of our method on multiple widely adopted benchmark datasets for node classification tasks.

2 Notations and Preliminaries

We let $G = (V, E)$ denote a graph, where $V = \{v_1, v_2, \dots, v_N\}$, $E \in V \times V$ are the set of nodes and edges of the graph, respectively. Each node in the graph has D -dimensional features, and the feature matrix can be denoted as $X = \{x_1, x_2, \dots, x_N\} \in \mathbb{R}^{N \times D}$. The adjacency matrix of a graph is: $A \in \{0, 1\}^{N \times N}$. In this work, we focus on undirected graphs, but the proposed method can be easily extended to directed graphs. Considering a set of above graphs $\mathcal{G}$, we aim to learn an encoder $f: \mathbb{R}^{N_i \times D} \times \{0, 1\}^{N_i \times N_i} \rightarrow \mathbb{R}^{N_i \times d}$, $d \ll D$ that maps the feature matrix X_i and adjacency matrix A_i in graph $G_i \in \mathcal{G}$ to low dimensional node representations $H_i \in \mathbb{R}^{N_i \times d}$, where H_i provides highly discriminative features in graph G_i for the downstream task. We associate each graph G_i with label $Y_i \in \mathcal{Y}$ in the task and assume (G_i, Y_i) are IID sampled from a global joint distribution $\mathbb{P}_{\mathcal{G} \times \mathcal{Y}} = \mathbb{P}_{\mathcal{Y}|\mathcal{G}}\mathbb{P}_{\mathcal{G}}$.

Federated Learning: Federated learning allows multiple clients to collaboratively train a global model without exposing private data from each client. Assume we have K clients with locally collected data inaccessible from others, i.e., $D_k = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{N_k}$, where N_k is the number of data instances. In general, the global model aims to minimize the global objective function $\mathcal{L}(\phi)$ for all clients as:

$$\min_{\phi} \mathcal{L}(\phi) = \sum_{k=1}^K \lambda_k \mathcal{L}_k(\phi_k), \quad (1)$$

where λ_k is the weight of the k -th client. The local objective function $\mathcal{L}_k(\phi_k)$ of client k is defined over local graph sets $\mathcal{D}_k$ as follows:

$$\mathcal{L}_k(\phi_k) \triangleq \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}_k} \ell(\phi_k; \mathbf{x}, \mathbf{y}), \quad (2)$$

where $\mathbf{x}$ is the input data with label $\mathbf{y}$ and $\ell(\cdot)$ represents the loss function, such as cross-entropy loss. Due to the distributed nature of customer data, the global objective function $\mathcal{L}(\phi)$ is optimized round by round. Specifically, in the r -th round of communication, each client locally trains their private data locally, thereby obtaining the optimization model $\{\phi_k^r\}_{k=1}^K$. These optimized models are then sent to a central server to derive a global model for the $(r+1)$ -th round of communication through the aggregation mechanism $\text{AGG}(\cdot)$ as follows:

$$\phi^{r+1} = \text{AGG}(\{\phi_k^r\}_{k=1}^K). \quad (3)$$

3 Methodology

In this section, we propose a novel framework personalized federated graph augmentation (PFGA) framework for efficient graph learning in federated scenarios. We first propose a centralized graph augmentation framework with graph regularization, which augments the graph samples with minimal sufficient information. We formulate the graph augmented framework as a min-max adversarial learning problem. We then propose the personalized federated graph augmentation framework with AVG augmentation and Clustered augmentation under the min-max adversarial learning.

3.1 Regularized Adversarial Graph Augmentation

Traditional adversarial graph augmentation like AD-GCL [24] is inspired by Graph Information Bottleneck (GIB) [30, 38] to learn the minimal information that is sufficient to identify each graph. Different from the supervised GIB that requires the label information from the downstream tasks, AD-GCL does not require the label and thus is suitable for self-supervised training where there exist no labels. It optimizes the min-max optimization problem as follows:

$$\text{AD-GCL: } \min_{\psi} \max_{\theta} \mathbb{E}_{G \sim \mathbb{P}_G} [I(f_{\theta}(G); f_{\theta}(G^+))], \quad (4)$$

where $G^+ \sim g_{\psi}(G)$, $f_{\theta}(\cdot)$ is the graph representation model and g_{ψ} is the graph generator. Without limiting the generation process, the augmented graph might drop the essential information related to the task and result in the collapse of training. To guarantee that the generated graph contains useful information, AD-GCL adds the constraint of dropping edges to ensure the generated graph is not too far from the original graph. The hidden assumption is that a slight change in the graph would not destroy the key information related to the label. To demonstrate the assumption clearly, we have the following definition.

DEFINITION 1. (Feasible Set) For a given graph $G \in \mathcal{G}$, we define the graph $\tilde{G} \in \mathcal{G}$ as its feasible sample if $\tilde{G}$ and G have the same label. Then the feasible set of graph G can be denoted as $\mathbb{D}_G = \{\tilde{G} \in \mathcal{G} | Y(G) = Y(\tilde{G})\}$.

The above feasible set requires the label information for the downstream tasks. For the supervised learning with the label information, the augmented graphs should be in the feasible set such that they can include the key information related to the label, i.e.,

$\tilde{G} \in \mathbb{D}_G$. For self-supervised learning without label information, we propose an approximate method via regularizing the generated graphs, i.e., $\tilde{\mathbb{D}}_G = \{\tilde{G} \in \mathcal{G} | d_g(G, \tilde{G}) \leq \epsilon\}$. Here, the distance function $d_g(G, \tilde{G})$ measures the difference between two graphs. The approximated feasible set $\tilde{\mathbb{D}}_G$ includes all graphs similar to original graph G . With the hidden assumption, the feasible set $\mathbb{D}_G$ can be approximated by $\tilde{\mathbb{D}}_G$. Then, when we adopt $\tilde{\mathbb{D}}_G$ to substitute the feasible set $\mathbb{D}_G$ for self-supervised learning, the augmented graph $\tilde{G}$ in the approximated feasible set i.e., $\tilde{G} \in \tilde{\mathbb{D}}_G$ also keeps the key information related to the label. To add the constraint for augmented graphs into the objective, we define the graph distance between a graph G and a graph set $\mathcal{D}$ as $D(G, \mathcal{D}) = \min_{G' \in \mathcal{D}} d_g(G, G')$. Then, we propose a novel general regularized adversarial graph augmentation (RAGA) method for both self-supervised and supervised augmentations as follows:

$$\text{RAGA: } \min_{\psi} \max_{\theta} \mathbb{E}_{G \sim \mathbb{P}_G} [I(f_{\theta}(G); f_{\theta}(\tilde{G})) + \lambda D(\tilde{G}, \mathbb{D}_G)], \quad (5)$$

where $\tilde{G} \sim g_{\psi}(G)$ and λ is positive constant. AD-GCL is a special case of RAGA for self-supervised augmentation, i.e., AD-GCL adopts the regularization of total dropped edges, i.e., $\epsilon = 0$ and $d_g(G, \tilde{G}) = \frac{1}{|E_G|} \sum_{e \in E_G} [1 - w_e^+]$, where w_e^+ is the probability of e in G^+ . Different from AD-GCL, RAGA is also suitable for supervised augmentation if the label information is available. The augmented samples in supervised cases should maximize the information related to the label equivalently, e.g., $D(G, \mathbb{D}_G) = -I(\tilde{G}, Y(G))$. The adversarial min optimization requires the generated graph samples to have minimal mutual information with the original graph but keep the key information related to the label, which guarantees the minimal sufficient information for generated samples. Different from AD-GCL which utilizes the GNN framework to learn augmented graphs, we instantiate the RAGA via designing a novel C-VGAE framework inspired by VGAE [10] to generate augmented graphs and a graph contrastive learning (GCL) framework with inter-GCL to capture the mutual information of original graphs and augmented graphs and intra-GCL to capture the information inside original graphs. We put the details of RAGA into Appendix A.

THEOREM 1. Denote the optimal encoder for RAGA as f^* . If $\lambda \geq 1$, then the encoder extracts the mutual information between the original graph and the label for both self-supervised and supervised augmentations, i.e., $I(f^*(G); Y) = I(G; Y)$.

Theorem 1 states that the optimal encoder RAGA for both self-supervised and supervised learning method can capture the most important information related to the label, although they extract such information from different views. It also shows that our RAGA provides a unified framework for both supervised GRL methods and self-supervised GRL methods with graph augmentations.

To demonstrate the essence of RAGA more clearly, we rewrite it into a min-max optimization:

$$\min_{\psi} \max_{\theta} R(\theta, \psi) = \mathbb{E}_{G \sim \mathbb{P}_G} [I(\theta, \psi; G) + \lambda h(\psi; G)], \quad (6)$$

where $I(\theta, \psi; G)$ represents the mutual information function and $h(\psi; G)$ represents the regularized constraint. Note that different ψ indicates different graph generators, which enhance the graph representation model by generating graph samples from different

views. To differentiate our graph generator from a traditional rule-based one, we rename our graph generator as the generative model. Based on the min-max optimization, we design the personalized federated graph augmentation framework. Our personalized federated learning framework is not limited to the graph augmentation, but the general min-max optimization under FL when only sharing the parameters of max optimization.

3.2 Personalized Federated Graph Augmentation

We separate the training of min-max optimization into two alternative problems. The first problem is to find the best graph representation given the generative model. The second problem is to find the best generative model given the GRL model. More formally, the alternative process can be demonstrated as follows:

$$\tilde{\theta} \leftarrow \arg \max_{\theta} \mathbb{E}_{G \sim \mathcal{P}_G} l(\theta, \tilde{\psi}; G), \quad (7)$$

$$\tilde{\psi} \leftarrow \arg \min_{\psi} \mathbb{E}_{G \sim \mathcal{P}_G} [l(\tilde{\theta}, \psi; G) + \lambda h(\psi; G)]. \quad (8)$$

The process of optimizing $\tilde{\theta}$ and $\tilde{\psi}$ is updated in an alternative way until converging to the optimal. Compared to simultaneous methods, the alternating one may be better in bypassing local optima. Two recent works [22, 31] have applied the federated learning method into the min-max optimization problem via updating the parameters of $\tilde{\theta}$ and $\tilde{\psi}$ in each clients to the server synchronously. They also provide theoretical analysis for their convergence. In this paper, we proposed a novel asynchronous federated learning framework for the min-max optimization problem that is more suitable for the federated graph augmentations. The overall framework of min-max optimization under asynchronous federated learning is demonstrated in Fig. 1.

Federated graph augmentation: The federated graph augmentation is based on the alternative training of min-max optimization in an asynchronous method. In the federated learning, each client i might have a different subgraph data set sampled from the global graphs, formulating different local joint distributions $\mathbb{P}_{\mathcal{G}_i \times \mathcal{Y}_i}$. We denote the subgraph sets for client i as $\mathcal{D}_i$. The dataset $\mathcal{D}_i$ is the private information for client i inability to share with other clients. Instead of sharing the parameters of GRL models, we share the parameters of the generative model. To achieve this goal, we should first estimate the parameters of the local GRL model for client i based on $\mathcal{D}_i$ as follows:

$$\theta_i^* = \arg \max_{\theta_i} \mathbb{E}_{G \in \mathcal{D}_i} l(\theta_i, \psi; G). \quad (9)$$

With the personalized parameters of GRL models θ_i^* for each client, we have the federated graph augmentation to learn the global optimal generative models as follows:

$$\min_{\psi} \{L(\psi) = \frac{1}{K} \sum_{i=1}^K L_i(\psi)\}, \quad (10)$$

$$L_i(\psi) = \mathbb{E}_{G \sim \mathcal{P}_G} [l(\theta_i^*, \psi; G) + \lambda h(\psi; G)], \quad (11)$$

where $L_i(\psi)$ is the local objective function under the globally generative model and optimal personalized GRL models and $L(\psi)$ is the global objective function.

PFGA with AVG augmentation: Inspired by the FedAvg framework, our personalized federated graph can also adopt the average (AVG) operator to aggregate the parameters of the generative model in each client. With the personalized parameters of GRL models θ_i^* , each client can learn the optimal parameters of a locally generative model ψ_i based on $\mathcal{D}_i$. Then, we can aggregate the local generative parameters ψ_i into global generative parameters ψ via an average operator. More formally, we have:

$$\psi = \frac{1}{K} \sum_{i=1}^K \psi_i, \text{ where } \psi_i = \arg \min \mathbb{E}_{G \in \mathcal{D}_i} [l(\theta_i^*, \psi_i; G) + \lambda h(\psi_i; G)]. \quad (12)$$

Different generative models generate graph samples from different views, which enrich the diversity of graph samples. With the assistance of graphs with different views, the local GRL models can prevent from the model over-fitting and have better model generalization. For each client, it also has the globally generative model to augment the graph sample from a global graph view, which may remedy the missing information due to missing edges across different clients. More formally, we can optimize the graph representation learning based on dataset $\mathcal{D}_i$ as follows:

$$\theta_i^* = \arg \max_{\theta_i} \mathbb{E}_{G \in \mathcal{D}_i} [l(\theta_i, \psi_i; G) + \beta l(\theta_i, \psi; G)], \quad (13)$$

where β balances the local augmentation and the global augmentation.

PFGA with clustered augmentation: In the AVG federated graph augmentation, each client not only contributes a local generative model but also receives a globally generative model. Note that one generative model can provide one view of graph augmentation and thus improves the diversity of graph samples. This diversity benefits the generalization ability of the model because the model is not limited to the data characteristics of a single client. If each client shares its locally generative model directly instead of by the globally generative model, the benefit from improved graph diversity would be the highest. However, each client should download all generative models, resulting in significant communication overhead.

To solve this issue, we proposed a clustered federated graph augmentation, which balances the diversity and the communication cost. We consider cluster set $C = \{c_1, \dots, c_M\}$, where M is the number of clusters and c_i includes the client index for cluster i . We attempt to put the clients with similar parameters into one cluster so as to avoid the communication cost. We adopt the cosine similarity to measure the similarity of locally generative models and have the following optimization:

$$\min_C \sum_{c_i \in C} \sum_{c_j \in C} \cos(\tilde{\psi}_i, \tilde{\psi}_j), \quad (14)$$

where $\tilde{\psi}_i = 1/|c_i| \sum_{j \in c_i} \psi_j$. As the larger cosine similarity indicates the closer distance, minimizing the objective is to find the best partition C where the parameters in one cluster are far from others.

To solve the partition problem, we propose a fast heuristic partition algorithm with details in Appendix B. We initialize each client as a single cluster and the server maintains a partition set of N clusters. We greedily decrease the value of Eq. 14 by merging the similar clusters at each round via a threshold τ until converging to the minimal value. Note that the calculation of similarity for

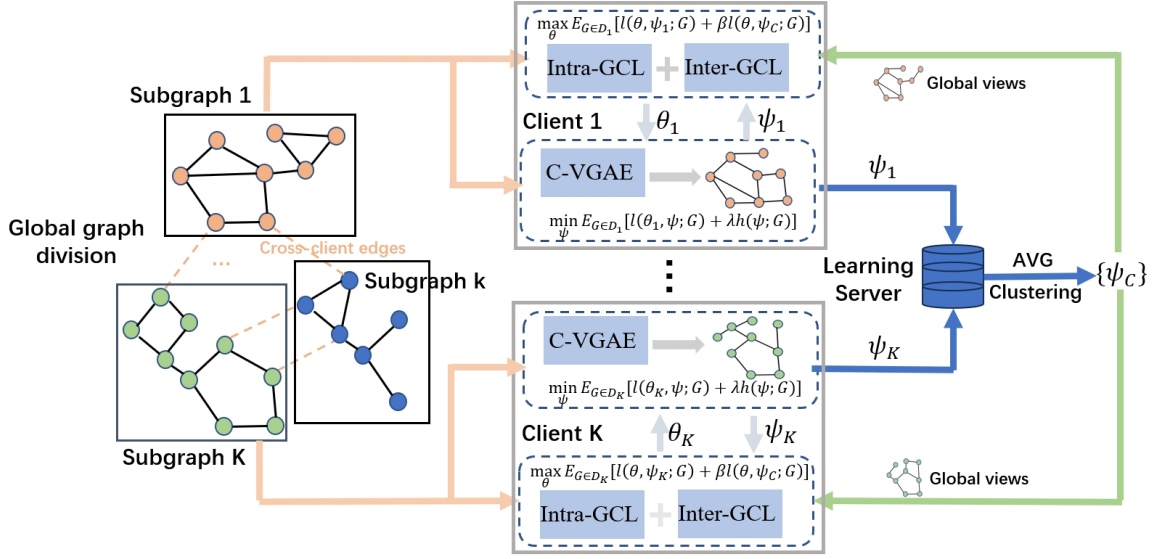


Figure 1: The overall framework of PFGA

two models is highly time-consuming. To accelerate the merging process, we introduce the anchor model by averaging all models. We calculate the similarity of each cluster model and anchor model instead of pair-wise cluster to cluster models. This reduces the similarity calculation from $O(N^2)$ to $O(N)$. Note that simplifying the calculation of similarity might scarify the accuracy of clustering.

With the cluster partition C , the server merges the generative models in each cluster and sends the clustered generative models to each client. Each client would have $|C|$ generative models to provide graph samples from $|C|$ views. Then, we can optimize graph representation learning as:

$$\theta_i^* = \arg \max_{\theta_i} \mathbb{E}_{G \in \mathcal{D}_i} [l(\theta_i, \psi_i; G) + \frac{\beta}{|C|} \sum_{c_j \in C} l(\theta_i, \tilde{\psi}_j; G)]. \quad (15)$$

Privacy and communication cost: For privacy, previous methods usually share the GRL models, which serve the tasks and may contain key information. For instance, if someone gets the GRL model for one client, it can infer the result/label of one new sample and mimic its decision. If he only gets the auxiliary model, it cannot infer any useful information related to its label. Since PFGA only shares auxiliary models instead of key GRL models, we have a lower privacy risk. For the communication cost, when we use the AVG to share the generative model, it does not include extra communication costs. Instead, since we utilize the graph generator to assist in the training of the local GRL model, it would be trained more sufficiently in each round and may result in a faster convergence rate verified in the experiment. In addition, our method provides the potential to design larger GRL models without increasing the communication burden.

4 Theoretical Analysis

In this subsection, we provide the convergence analysis for the min-max optimization under the asynchronous federated learning

framework. Before the analysis, we first introduce two assumptions over the objective function of min-max optimization $R(\psi, \theta)$.

ASSUMPTION 1. $R(\psi, \theta)$ is continuously differentiable and l -smooth on $\Psi \times \Theta$, i.e., $[\nabla R(\psi, \theta), \nabla R(\psi, \theta)]$ is l -Lipschitz on $\Psi \times \Theta$ and for every $\psi, \psi' \in \Psi, \theta, \theta' \in \Theta$ we have

$$\begin{aligned} & \|\nabla R(\psi, \theta) - \nabla R(\psi', \theta')\|_2^2 + \|\nabla R(\psi, \theta) - \nabla R(\psi', \theta')\|_2^2 \\ & \leq l^2 (\|\psi - \psi'\|_2^2 + \|\theta - \theta'\|_2^2). \end{aligned} \quad (16)$$

ASSUMPTION 2. The variance of stochastic gradients for $R(\psi, \theta)$ is bounded over D if for every $(\psi, \theta) \in \Psi \times \Theta$,

$$\begin{aligned} & \mathbb{E}_{g \in D} [\|\nabla R_\theta(\psi, \theta; g) - \nabla R_\theta(\psi, \theta; D)\|_2^2] \leq \sigma_\theta^2, \\ & \mathbb{E}_{g \in D} [\|\nabla R_\psi(\psi, \theta; g) - \nabla R_\psi(\psi, \theta; D)\|_2^2] \leq \sigma_\psi^2. \end{aligned} \quad (17)$$

Assumption 1 about L -smoothness is standard for the convergence analysis. For instance, pFedMe[25] provides such an assumption over the local objective function. In our min-max optimization, we provide this assumption over parameters of GRL models and generative models. Assumption 2 is widely used in federated learning in which the sampling noise of the client's data distribution is quantified and upper bounded.

When we analyze the convergence of min-max optimization under federated learning, we separate this problem as a bi-level optimization problem. We optimize the max problem for GRL models over θ and analyze the convergence of the min problem under federated learning. With the decoupling of min-max optimization, the maximized objective has the following characteristics. The proofs of theorems and lemmas can be found in [40].

LEMMA 1. If the function $R(\psi, \theta)$ is a μ_θ -strongly-concave function over Θ and Assumption 1 is satisfied, then the maximized objective $R_m(\psi) := \max_{\theta} R(\psi, \theta)$ is $(l + l^2/2\mu_\theta)$ -smooth, i.e., for every, $\psi_1, \psi_2 \in \Psi$ it satisfies

$$\|\nabla R_m(\psi_2) - \nabla R_m(\psi_1)\|_2 \leq (l + l^2/2\mu_\theta) \|\psi_2 - \psi_1\|_2. \quad (18)$$

Lemma 1 states that the μ_θ -strongly-concave function over Θ guarantees the smoothness of the maximized objective $R_m(\psi)$, which provides the foundation of convergence under federated learning. However, it is not straightforward to obtain optimal parameters of GRL models θ in closed-form. Instead, we use an iterative first-order approach to obtain an approximate $\tilde{\theta}$ with estimated errors upper bounded by δ_θ . When we update the objective, we usually require the gradient $\nabla R(\psi, \theta)$. We provide the unbiased estimate of $\nabla R(\psi_t, \tilde{\theta}_i)$ by sampling a mini-batch of dataset D_i ,

$$\tilde{\nabla} R(\psi_t, \tilde{\theta}_i) = \frac{1}{|D_i|} \sum_{g \in D_i} \nabla R_\theta(\psi, \tilde{\theta}_i; g), \quad (19)$$

where $\tilde{\theta}_i$ is a estimation of optimal θ_i^* under dataset D_i

DEFINITION 2. *SGD update:* $\psi_{t+1} = \psi_t - \eta_\psi \nabla \tilde{R}(\psi_t, \tilde{\theta}_i)$ for client i , where η_ψ is the adaptive step size.

In the SGD update, we only update the generative model with the estimated optimal parameters of GRL models. With the SGD update, the min-max optimization under FL is reduced to the traditional minimization problem under FL. For the convex and concave objective functions, we have the convergence analysis in the following theorem.

THEOREM 2. *Consider that $R(\psi, \theta)$ is a μ_ψ -strongly-convex and μ_θ -strongly-concave function over $\Psi \times \Theta$, and Assumption 1 and 2 are satisfied. If $T \geq \frac{2}{\mu_\psi \tilde{\eta}}$, there exist an $\eta_\psi \leq \tilde{\eta}$ where $\tilde{\eta} = \frac{1}{6(2l+l^2/\mu_\theta)}$ such that the gap of objective value under SGD updates is bounded by*

$$L(\tilde{\psi}_i^T) - L(\psi^*) \leq \mu_\psi e^{-\eta_\psi T/2} \Delta_0 + C_1 \sigma_R^2 + C_2 \delta_\theta^2 + C_3 \sigma_\psi^2, \quad (20)$$

where $\Delta_0 = \|\psi_0 - \psi^*\|_2^2$, $\sigma_R^2 = \frac{1}{K} \sum_{i=1}^K \|\nabla L_i(\psi^*)\|_2^2$, $\tilde{\psi}_i^T := \sum_{t=0}^{T-1} \alpha_t \psi_t / A_T$ with $\alpha_t = (1 - \eta_\psi \mu_\psi / 2)$ and $A_T = \sum_{t=0}^{T-1} \alpha_t$, $C_1 = 6\eta_\psi$ and $C_2 = (2l^2/\mu_\psi + 3\eta_\psi l^2)$ and $C_3 = (4/\mu_\psi + 3l^2\eta_\psi)/|D|$.

Theorem 2 demonstrates the convergence of the globally generative model with four error terms. The first term shows that a carefully chosen constant step size can reduce the initial error linearly. The second term is due to client drift with multiple local updates, which would also be reduced linearly. The third item is the estimated error over θ and the fourth is the variance in estimation over ψ . The total gap is upper bounded by $O(\frac{2l^2}{\mu_\psi} \delta^2 + \frac{4}{\mu_\psi |D|} \sigma_\psi^2)$. As the increase of graph dataset, the error in the fourth item would be ignored and the upper bound is further reduced to $O(\frac{2l^2 \delta^2}{\mu_\psi})$.

THEOREM 3. *Consider that the function $R(\psi, \theta)$ is non-convex over Ψ but μ_θ -strongly-concave over Θ , and Assumption 1 and 2 are satisfied. If $\eta \leq \tilde{\eta}$ where $\tilde{\eta} = \frac{1}{3(2l+l^2/\mu_\theta)}$ such that the gradients under SGD updates are bounded by*

$$\frac{1}{2T} \sum_{t=0}^{T-1} E[\|\nabla L(\psi_t)\|_2^2] \leq \frac{2\Delta}{\eta_\psi T} + (2 + 3l\tilde{\eta}_\psi)(l^2 \delta_\theta^2 + \sigma_\psi^2 / |D|), \quad (21)$$

where $\Delta := E[L(\psi_0) - L(\psi_T)]$, $\tilde{l} = l + \frac{l^2}{2\mu_\theta}$.

Theorem 3 demonstrates the convergence of the accumulated gradients with three error terms for the non-convex and concave objective functions. A carefully chosen constant step size can reduce the initial error linearly in the first term. The second item

is the estimated error over θ and the third item is the variance in estimation over ψ . The accumulated gradient is upper bounded by $O(2l^2 \delta^2 + \frac{2}{|D|} \sigma_\psi^2)$. As the increase of dataset, the error in the second item would be ignored and the upper bound is reduced to $O(2l^2 \delta^2)$.

For the non-convex and non-concave objective function, we cannot leverage the characteristics of maximized objective as in Lemma 1. To provide the convergence analysis, we utilize the general l -th smoothness in Assumption 1 and update the gradient in an alternative way, i.e., alternative SGD is shown as the following definition.

DEFINITION 3. *A-SGD update:* $\psi_{t+1} = \psi_t - \eta_\psi \nabla \tilde{R}_\psi(\psi_t, \theta_t)$ and $\theta_{t+1} = \theta_t + \eta_\theta \tilde{\nabla} R_\theta(\psi_{t+1}, \theta_t)$, where η_ψ and η_θ are the step sizes.

The A-SGD update is from the alternative process in Eq. 9 and Eq. 10. When we update their objectives, we usually require the gradients $\nabla R_\psi(\psi_t, \theta_t)$ and $\nabla R_\theta(\psi_{t+1}, \theta_t)$, which are unbiased estimated by sampling a mini-batch of dataset D_i as follows:

$$\begin{aligned} \tilde{\nabla} R_\psi(\psi_t, \theta_t) &= \frac{1}{|D_i|} \sum_{g \in D_i} \nabla R_\psi(\psi_t, \theta_t; g), \\ \tilde{\nabla} R_\theta(\psi_{t+1}, \theta_t) &= \frac{1}{|D_i|} \sum_{g \in D_i} \nabla R_\theta(\psi_t, \theta_t; g). \end{aligned} \quad (22)$$

THEOREM 4. *Consider that the function $R(\psi, \theta)$ is non-convex and non-concave over $\Psi \times \Theta$, and Assumptions 1 and 2 are satisfied. If $\max\{\eta_\theta, \eta_\psi\} \leq \frac{1}{4l}$ such that the gradients under A-SGD updates are bounded by*

$$\begin{aligned} &\frac{1}{T} \sum_{t=0}^{T-1} [E[\|\nabla R_\psi(\psi_t, \theta_t)\|_2^2 + \|\nabla R_\theta(\psi_{t+1}, \theta_t)\|_2^2]] \\ &\leq \frac{\Delta_R}{\min\{\eta_\psi, \eta_\theta\} T} + \gamma C_1 \sigma_\theta^2 + \gamma C_2 \sigma_\psi^2 \end{aligned} \quad (23)$$

where $\Delta_R = E[R(\psi_0, \theta_0) - R(\psi_T, \theta_T)]$, $C_1 = \frac{1+l\eta_\theta}{|D|}$ and $C_2 = \frac{1+l\eta_\psi}{|D|}$ and $\gamma = \frac{\max\{\eta_\psi, \eta_\theta\}}{\min\{\eta_\psi, \eta_\theta\}}$.

Theorem 4 demonstrates the convergence of the accumulated gradients with three error terms for the non-convex and non-concave objective functions. The initial error in the first term can be reduced linearly. The second and third items are the variance in estimation over θ and ψ . The accumulated gradient is upper bounded in average to $O(\frac{\gamma}{|D|} \sigma_\theta^2 + \frac{\gamma}{|D|} \sigma_\psi^2)$. As the increase of graph dataset, the error can be ignored and the optimal gradients would converge to zero.

5 Experiments

We first analyze the overall performance with non-overlapping graphs via random division and overlapping cases via community division. We then provide in-depth analysis including complementarity analysis, convergence analysis, ablation study and visual node representation. We also provide the performance analysis under extremely imbalanced division and in-depth experiments with parameter sensitivity analysis in Appendix C.

5.1 Experimental Setups

Datasets: We evaluate our PFGA in node classification tasks on six benchmark datasets: citation networks (Cora, CiteSeer [20]), co-purchasing relationship networks (Amazon-Computers and Amazon-Photo [23]) and Coauthor Academic Network (Coauthor-CS [23]).

Baselines: We compare six baselines, including FedAvg [19], FedProx [15], FedPub [1], pFedHN [21], FedSage+ [39], FedGCN [35] with two variants of our method, i.e., PFGA with AVG augmentation (PFGA-A) and PFGA with clustered augmentation (PFGA-C).

Implementation Details: The data is split into training, validation, and test sets in a 60/20/20% ratio, except for the ogbn-arxiv dataset, which is split in a 10/45/45% ratio due to its larger graph size. For the non-overlapping case, we divide the clients into 5, 10 and 20 groups via random division to avoid overlapping nodes across clients. For the overlapping scenario, we set the same split ratio as the non-overlapping case. We divide the clients into 10, 30 and 50 groups via Metis library function [9], which keeps the local structure via community division. More specifically, we first use the METIS algorithm to divide the original graph into 2, 6, and 10 disjoint sub-graphs, corresponding to 10, 30 and 50 clients respectively. Then, in each split subgraph, we randomly sample half of the nodes and their associated edges and use them as client-specific subgraphs. This process is performed five times, resulting in five distinct but overlapping subgraphs, each obtained from METIS. We adopt the SGD optimizer with a learning rate of 0.1 and the micro-F1 score as the evaluation metric. We carry out experiments via a NVIDIA A100 with 80GB.

5.2 Experimental Results

Overall Comparisons: In the non-overlapping scenario, Table 1 shows that our PFGA method outperforms all the baselines with different clients across all datasets significantly. In addition, PFGA-C is slightly better than PFGA-A even though we sacrifice lots of its performance due to the compromise of huge computation costs in clustering. Notably, as the number of clients increases, the performance of all baselines decreases significantly in most datasets. For instance, the state-of-the-art method pFedHN decreased from 87.2 to 83.8, i.e., a 3.4% reduction, when the number of clients increased from 5 to 20 in Amazon-Computer. The reason behind this is that the larger number of clients usually indicates less graph data in each client, resulting in more serious model over-fitting. However, our PFGA method demonstrates robust performance, e.g., it decreases from 87.3 to 86.7, i.e., only 0.6% reduction in Amazon-Computer. Our PFGA approach allows each client’s model to learn more comprehensive parameters via augmented graph samples from generative models, avoiding model over-fitting to local graph data.

In the overlapping scenario, Table 2 illustrates that PFGA demonstrates significant performance improvements over all baselines across all datasets. Notably, local models performed better in community division than random division, underscoring the importance of local structure. For the Amazon-Photo and Amazon-Computers datasets, local models outperform certain federated learning baselines, indicating that sharing the parameters might result in negative impacts on graph representation learning. However, PFGA

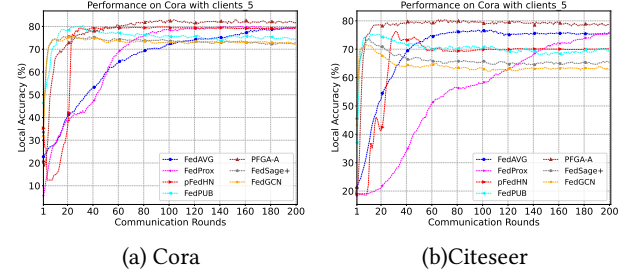


Figure 2: Convergence analysis

shares the parameters of the generative model, instead of the parameters of the GRL model and thus still demonstrates excellent performance with community division. Similar to the non-overlapping cases with random division, PFGA also demonstrates robust performance as the number of clients increases in the overlapping scenario.

Convergence Analysis: We provide convergence analysis for non-overlapping node scenarios with 5 clients under dataset Cora and CiteSeer in Figure 2. It demonstrates that our PFGA converges very quickly compared to the baselines. For instance, our PFGA-A increases from 20 to 83.2 after 100 communication rounds while FedAvg only increases from 22 to 71 under Cora with 5 clients. Since we utilize the graph generator to assist in the training of the local GRL model, the local GRL model would be trained more sufficiently in each global round and thus result in a faster convergence rate.

Ablation Study: We consider three ablation baselines and two alternative baselines for the non-overlapping case with 5 clients over three datasets. In the ablation baselines, we consider 'w/o FED&EM' with only GRL but no FL and generative model, 'w/o FED' with both GRL models and generative models but no FL, 'w/o EM' with only GRL under FL but no generative model. In the alternative baselines, we consider 'Fed GRL' which only shares GRL models instead of generative models, and 'Fed GRL&EM' which shares both GRL models and generative models.

Table 3 shows that the FL and the generative model play an important role in our framework. Our generative model can improve performance even without FL. For instance, our generative model improve the performance from 72.6 to 74.5, i.e., 1.9% improvement in Cora, when no federated learning is adopted. However, with FL, the generative model can capture global information from other clients, further increasing its advantage, i.e., from 74.5 to 83.2. In addition, both sharing all the parameters and sharing the parameters of GRL result in the performance reduction in all three datasets. The main reason is that penalization is a very important factor that impacts the performance of FL.

Complementarity Analysis: To demonstrate that our method indeed alleviates the complementarity problem, we add a baseline "Global_drop" in Table 3, which trains a centralized model via the global graphs but discarding missing global edges across clients. It does not have the heterogeneity problem but faces the complementarity problem. Most of the baselines have worse performance than "Global_drop" as they suffer from both heterogeneity and complementarity problems. For instance, the baseline "Fed. GRLEM"

Table 1: Overall comparisons for node classification tasks on the non-overlapping node scenario. (mean $\pm$ std. over 10 runs).

Algorithm	Cora			CiteSeer			Amazon-Computer		
	5 Clients	10 Clients	20 Clients	5 Clients	10 Clients	20 Clients	5 Clients	10 Clients	20 Clients
Local	72.6 $\pm$ 1.12	58.2 $\pm$ 1.82	48.3 $\pm$ 2.64	67.5 $\pm$ 1.03	62.4 $\pm$ 1.51	55.0 $\pm$ 2.83	83.4 $\pm$ 0.34	82.0 $\pm$ 0.27	77.8 $\pm$ 0.56
FedAVG	79.7 $\pm$ 0.78	76.2 $\pm$ 0.94	71.8 $\pm$ 0.93	75.2 $\pm$ 0.93	72.5 $\pm$ 1.03	72.7 $\pm$ 1.05	83.5 $\pm$ 1.34	77.9 $\pm$ 0.41	71.7 $\pm$ 0.53
FedProx	80.7 $\pm$ 0.76	76.2 $\pm$ 0.94	74.7 $\pm$ 1.20	76.0 $\pm$ 0.88	72.5 $\pm$ 1.03	72.2 $\pm$ 1.47	81.8 $\pm$ 0.33	85.2 $\pm$ 0.51	83.5 $\pm$ 0.73
FedPUB	79.8 $\pm$ 1.06	73.5 $\pm$ 0.82	73.5 $\pm$ 0.87	76.6 $\pm$ 0.96	72.7 $\pm$ 0.64	72.1 $\pm$ 0.71	83.0 $\pm$ 1.54	85.6 $\pm$ 2.06	81.5 $\pm$ 2.78
pFedHN	80.1 $\pm$ 0.41	75.0 $\pm$ 0.63	73.6 $\pm$ 0.41	76.3 $\pm$ 0.63	73.6 $\pm$ 0.82	73.5 $\pm$ 0.54	87.2 $\pm$ 0.24	86.0 $\pm$ 0.22	83.8 $\pm$ 0.69
FedSage+	76.7 $\pm$ 0.61	73.3 $\pm$ 0.82	73.3 $\pm$ 0.51	73.3 $\pm$ 0.56	72.5 $\pm$ 0.71	71.4 $\pm$ 0.53	80.7 $\pm$ 0.91	81.2 $\pm$ 0.27	82.5 $\pm$ 1.13
FedGCN	76.3 $\pm$ 0.64	74.3 $\pm$ 0.61	73.0 $\pm$ 0.44	71.7 $\pm$ 0.43	71.8 $\pm$ 0.51	71.7 $\pm$ 0.45	81.1 $\pm$ 1.13	82.0 $\pm$ 1.02	82.3 $\pm$ 0.81
PFGA-A	83.2 $\pm$ 0.28	80.1 $\pm$ 0.98	77.5 $\pm$ 0.38	78.0 $\pm$ 0.73	75.5 $\pm$ 1.08	76.9 $\pm$ 1.81	87.5$\pm$0.75	87.3 $\pm$ 1.32	86.4 $\pm$ 1.02
PFGA-C	83.6$\pm$0.25	80.2$\pm$0.31	78.3$\pm$0.44	78.8$\pm$0.91	76.3$\pm$0.99	77.5$\pm$1.31	87.3 $\pm$ 0.97	87.5$\pm$1.13	86.7$\pm$1.22

Algorithm	Coauthor-CS			Amazon-Photo			ogbn-arxiv		
	5 Clients	10 Clients	20 Clients	5 Clients	10 Clients	20 Clients	5 Clients	10 Clients	20 Clients
Local	89.4 $\pm$ 0.28	88.0 $\pm$ 0.36	86.8 $\pm$ 0.42	91.8 $\pm$ 0.19	89.6 $\pm$ 0.21	86.3 $\pm$ 0.75	45.0 $\pm$ 0.24	43.3 $\pm$ 0.14	41.1 $\pm$ 0.16
FedAVG	91.3 $\pm$ 0.28	90.5 $\pm$ 0.28	91.1 $\pm$ 0.32	89.5 $\pm$ 2.22	88.2 $\pm$ 0.11	85.9 $\pm$ 0.23	43.9 $\pm$ 0.43	44.4 $\pm$ 0.26	42.7 $\pm$ 0.33
FedProx	90.9 $\pm$ 0.23	90.2 $\pm$ 0.33	91.2 $\pm$ 0.41	91.3 $\pm$ 0.26	92.1 $\pm$ 0.15	89.8 $\pm$ 0.19	47.5 $\pm$ 0.30	48.0 $\pm$ 0.22	47.6 $\pm$ 0.10
FedPUB	91.3 $\pm$ 0.19	90.0 $\pm$ 0.15	90.6 $\pm$ 0.22	90.5 $\pm$ 1.52	86.1 $\pm$ 1.89	87.5 $\pm$ 1.99	45.2 $\pm$ 0.59	46.1 $\pm$ 0.33	43.7 $\pm$ 0.28
pFedHN	90.8 $\pm$ 0.27	89.8 $\pm$ 0.96	90.8 $\pm$ 0.63	93.2 $\pm$ 0.23	92.8 $\pm$ 0.29	91.5 $\pm$ 0.75	49.1 $\pm$ 0.15	48.3 $\pm$ 0.18	48.4 $\pm$ 0.22
FedSage+	91.9 $\pm$ 0.29	90.8 $\pm$ 0.11	90.9 $\pm$ 0.28	92.9 $\pm$ 0.65	92.8 $\pm$ 0.44	92.9 $\pm$ 0.47	49.9 $\pm$ 0.22	49.9 $\pm$ 0.23	49.9 $\pm$ 0.14
FedGCN	91.8 $\pm$ 0.11	90.8 $\pm$ 0.23	90.7 $\pm$ 0.16	92.8 $\pm$ 0.42	92.5 $\pm$ 0.16	92.4 $\pm$ 1.02	49.5 $\pm$ 0.13	49.8 $\pm$ 0.15	49.9 $\pm$ 0.17
PFGA-A	92.4 $\pm$ 0.56	91.4 $\pm$ 0.32	91.4$\pm$0.49	93.2 $\pm$ 0.22	93.4$\pm$0.27	93.0 $\pm$ 0.69	50.2$\pm$0.28	51.0 $\pm$ 0.33	50.6 $\pm$ 0.41
PFGA-C	92.6$\pm$0.47	91.5$\pm$0.35	91.3 $\pm$ 0.71	93.5$\pm$0.98	93.3 $\pm$ 0.45	93.3$\pm$0.33	50.1 $\pm$ 0.31	51.3$\pm$0.43	51.0$\pm$0.29

Table 2: Overall comparisons for node classification tasks on the overlapping node scenario. (mean $\pm$ std. over 10 runs).

Algorithm	Cora			CiteSeer			Amazon-Computer		
	10 Clients	30 Clients	50 Clients	10 Clients	30 Clients	50 Clients	10 Clients	30 Clients	50 Clients
Local	77.6 $\pm$ 0.32	72.9 $\pm$ 0.55	74.4 $\pm$ 0.46	72.6 $\pm$ 0.41	67.3 $\pm$ 0.47	66.4 $\pm$ 0.53	89.1 $\pm$ 0.14	88.2 $\pm$ 0.21	88.3 $\pm$ 0.18
FedAVG	82.4 $\pm$ 0.28	77.7 $\pm$ 0.32	75.1 $\pm$ 0.45	76.1 $\pm$ 0.62	74.6 $\pm$ 0.71	73.6 $\pm$ 0.61	73.8 $\pm$ 3.02	71.2 $\pm$ 5.32	67.5 $\pm$ 4.12
FedProx	83.2 $\pm$ 0.38	80.8 $\pm$ 0.46	80.0 $\pm$ 1.21	77.2 $\pm$ 0.52	74.9 $\pm$ 0.63	74.4 $\pm$ 0.59	89.5 $\pm$ 0.19	87.6 $\pm$ 0.42	87.5 $\pm$ 0.55
FedPUB	82.6 $\pm$ 0.71	77.5 $\pm$ 0.81	74.4 $\pm$ 0.96	77.0 $\pm$ 0.35	74.7 $\pm$ 0.43	73.3 $\pm$ 0.67	79.6 $\pm$ 5.68	76.4 $\pm$ 5.23	74.9 $\pm$ 3.78
pFedHN	82.4 $\pm$ 0.78	77.9 $\pm$ 0.79	76.7 $\pm$ 0.85	77.9 $\pm$ 0.44	73.1 $\pm$ 0.28	72.9 $\pm$ 0.35	89.2 $\pm$ 0.11	89.1 $\pm$ 0.76	89.3 $\pm$ 0.65
PFGA-A	85.0 $\pm$ 0.67	83.0 $\pm$ 0.31	82.7 $\pm$ 0.17	80.3$\pm$0.43	77.6 $\pm$ 0.98	76.7 $\pm$ 0.76	90.5$\pm$0.65	90.1 $\pm$ 0.23	90.1$\pm$0.25
PFGA-C	85.3$\pm$0.88	83.2$\pm$0.35	83.2$\pm$0.91	80.1 $\pm$ 0.35	77.8$\pm$0.81	77.0$\pm$1.65	90.3 $\pm$ 0.43	90.2$\pm$0.51	90.1 $\pm$ 0.46

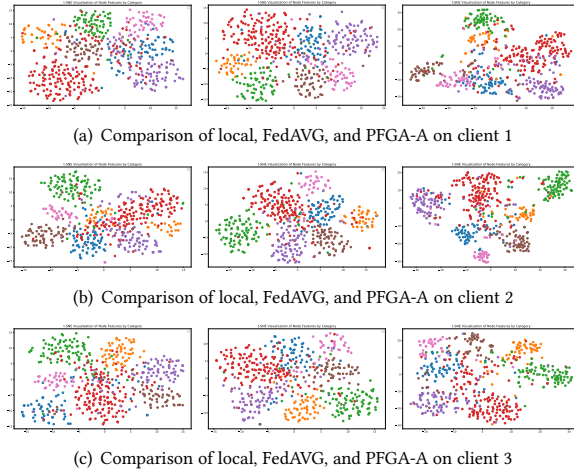
Algorithm	Coauthor-CS			Amazon-Photo			ogbn-arxiv		
	10 Clients	30 Clients	50 Clients	10 Clients	30 Clients	50 Clients	10 Clients	30 Clients	50 Clients
Local	91.6 $\pm$ 0.09	90.4 $\pm$ 0.13	89.8 $\pm$ 0.15	93.9 $\pm$ 0.18	92.3 $\pm$ 0.20	92.0 $\pm$ 0.17	49.1 $\pm$ 0.11	48.9 $\pm$ 0.11	48.0 $\pm$ 0.07
FedAVG	92.1 $\pm$ 0.22	89.9 $\pm$ 0.19	87.8 $\pm$ 0.81	89.2 $\pm$ 1.78	83.4 $\pm$ 3.21	80.2 $\pm$ 3.67	40.1 $\pm$ 0.91	42.8 $\pm$ 0.29	42.4 $\pm$ 0.46
FedProx	92.8 $\pm$ 0.09	92.5 $\pm$ 0.05	92.2 $\pm$ 0.07	94.1 $\pm$ 0.19	93.6 $\pm$ 0.15	92.6 $\pm$ 0.19	50.2 $\pm$ 0.24	48.8 $\pm$ 0.24	48.6 $\pm$ 0.17
FedPUB	92.2 $\pm$ 0.15	89.8 $\pm$ 0.54	88.1 $\pm$ 0.32	89.8 $\pm$ 2.53	81.4 $\pm$ 4.15	79.3 $\pm$ 4.54	46.7 $\pm$ 0.94	45.8 $\pm$ 0.24	44.5 $\pm$ 0.70
pFedHN	92.2 $\pm$ 0.10	91.5 $\pm$ 0.25	91.2 $\pm$ 0.29	93.9 $\pm$ 0.08	93.8 $\pm$ 0.08	93.6 $\pm$ 0.05	50.7 $\pm$ 0.16	50.8 $\pm$ 0.18	50.4 $\pm$ 0.15
PFGA-A	92.9$\pm$0.21	92.8 $\pm$ 0.53	92.7 $\pm$ 0.83	94.7 $\pm$ 0.77	94.2 $\pm$ 0.66	94.0 $\pm$ 0.47	51.7 $\pm$ 0.11	53.6 $\pm$ 0.21	53.8 $\pm$ 0.35
PFGA-C	92.8 $\pm$ 0.25	92.9$\pm$0.36	92.9$\pm$0.95	94.8$\pm$0.25	94.3$\pm$0.29	94.3$\pm$0.76	51.8$\pm$0.15	53.7$\pm$0.43	53.9$\pm$0.51

Table 3: Ablation Study for node classification

Algorithm	Cora	Citeseer	Photo
Global_Drop	80.8±0.69	77.0±0.97	92.7±0.21
w/o FED&EM	72.6±1.12	67.5±2.15	91.8±0.29
w/o FED	74.5±1.32	70.3±2.32	91.5±0.31
w/o EM	79.7±1.16	75.2±1.54	89.5±0.14
Fed. GRL	81.1±1.18	75.8±1.57	90.7±0.27
Fed. GRL&EM	81.4±1.41	76.9±1.62	92.4±0.26
PFGA-A	83.2±0.28	78.0±0.73	93.2±0.22

suffers from the heterogeneity problem since there does not exist personalized parameters while the baseline "Fed. GRL" suffers from the complementarity problem since the generative model can only augment the graph samples from local view. Our method outperforms the baseline "Global_drop" in most data sets, indicating that it remedies the missing global information via augmented graphs from globally generative models.

Visual Node Representation: We first train three GNN encoders under Local, FedAVG, and our proposed method and use t-SNE to visualize the learned representations. Specifically, we examine the node visualizations for three out of the five clients on the Cora dataset. As shown in Figure 3, compared to the other methods, our approach results in representations that form tighter clusters within each category, while maintaining greater separation between different categories. This leads to better classification of the nodes.

**Figure 3: t-SNE of node representation on Cora dataset**

6 Related Work

Federated Learning: Federated learning (FL)[15] is a distributed machine learning paradigm that enables multiple clients to collaboratively train models without sharing their data. This approach reduces reliance on data center resources and enhances data privacy by updating models locally and aggregating these updates on a central server. However, FL models often perform poorly when faced with severe data heterogeneity (non-independent and identically

distributed data)[8, 14]. To address data heterogeneity, advanced research has introduced various learning objectives that calibrate the update direction of local training away from the global model. Notable methods include FedProx [15], FedIR [6], SCAFFOLD [8], and MOON [13]. Additionally, model aggregation schemes such as FedAvgM [5], FedNova [28], FedMA [27], and FedBN [16] have shown efficacy in mitigating heterogeneity. Another promising direction is information-sharing strategies, which focus on synthesizing and sharing client information to alleviate heterogeneity [17, 41]. While the effectiveness of federated learning has been demonstrated, the challenges of data heterogeneity and personalization across clients remain inadequately addressed.

Graph Federated Learning: Recent studies have demonstrated the use of the federated learning (FL) framework to collaboratively train graph neural networks (GNNs) without sharing graph data [3, 18]. These methods can be broadly categorized into subgraph-level methods and graph-level methods. Graph-level FL methods assume that different clients possess completely disjoint graphs (e.g., molecular graphs), with some recent work [4, 26] addressing the non-i.i.d. nature of graph heterogeneity (i.e., differences in graph labels between clients). Subgraph-level FL, on the other hand, faces the unique challenge of missing but possible links between subgraphs, as these subgraphs are part of a larger global graph. To address this issue, existing methods enhance node information by requesting details from other subgraph nodes and connecting existing nodes with these enhanced nodes [34, 39]. However, this approach may compromise data privacy restrictions. Unlike these methods that focus on the missing link problem, this study addresses the issue by learning min-max adversarial graph augmentation under the federated framework."

7 Conclusion

In this work, we proposed a novel personalized federated graph augmentation framework, which consists of generative models and GRL models. We aggregated the locally generative model instead of the GRL model into the global models to alleviate the heterogeneity problem and utilized the globally generative model to generate samples from the global view to alleviate the complementarity problem. We formulated the training of the augmented and GRL models as a min-max adversarial optimization and provided its theoretical convergence analysis under FL. We apply our method limited to node classification tasks, and would extend it to other tasks beyond graphs in future works.

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A Instantiation of RAGA

Graph augmentation: To generate the positive graphs, we adopt a Variational Graph Auto-encoder (VGAE) [10] framework. VGAE is designed to learn low-dimensional representations of graphs in an unsupervised manner. It uses a probabilistic encoder-decoder architecture, with the encoder network learning a distribution over latent variables that encode the input graph and the decoder network reconstructing the input graph via latent variables. Our target is to generate the graph G^+ via the VGAE framework. Traditional VGAE utilizes $q(Z|G)$ to encode the graph and $p(G^*|Z)$ to decode the hidden embedding. In this subsection, we propose a novel Conditional VGAE (C-VGAE) that considers an extra conditional graph. The encoders learn a distribution over latent variables Z based on one rule-based positive graphs $\tilde{G}$ and the original graph G . The latent variables can be learned independently for different nodes:

$$q^s(Z^+ | G, \tilde{G}) = \prod_{i=1}^N q^+(z_i^+ | G, \tilde{G}). \quad (24)$$

Specifically, we assume a Gaussian distribution for the latent variable of each node:

$$q^+(z_i^+ | G, \tilde{G}) = \mathcal{N}(z_i^+ | [\mu_i, \tilde{\mu}_i], \text{diag}(\sigma_i^2, \tilde{\sigma}_i^2)), \quad (25)$$

where (μ_i, σ_i) are the mean and variance for node i in conditional graph G learned by $\mu = \text{GCN}_\mu(G)$, and $\log \sigma = \text{GCN}_\sigma(G)$ respectively. $(\tilde{\mu}_i, \tilde{\sigma}_i)$ are the mean and the variance for node i in the rule-based positive graph learned by $\tilde{\mu} = \text{GCN}_{\tilde{\mu}}(\tilde{G})$ and $\log \tilde{\sigma} = \text{GCN}_{\tilde{\sigma}}(\tilde{G})$ respectively.

The decoder calculates the probability of edges to reconstruct the graph based on the latent variables of node pairs shown as follows:

$$p(G^+ | Z) = \prod_{i=1}^N \prod_{j=1}^N p^+(A_{ij}^+ | z_i^+, z_j^+), \quad (26)$$

where $p(A_{ij} = 1 | z_i, z_j) = f_G(z_i, z_j)$, $A_{ij}^+ = 1$ indicates the existence of an edge between nodes i and j for the generated positive graph, and $f_G = \sigma(z_i^T, z_j)$ generates the probability of this edge.

The loss function includes the distance measurement between the generated graph and the original graph and the divergence of the node representation distribution and the normal distribution:

$$\mathcal{L}_V = E_{q(Z^+|G, \tilde{G})} [\log p^+(G^+ | Z^+)] - \gamma \text{KL}[q^+(Z^+ | G, \tilde{G}) \| p(Z)]. \quad (27)$$

The first item is the reconstruction loss with the cross-entropy function while the second term is the KL divergence between the variational and prior distributions. We utilize the isotropic Gaussian distribution $p(Z) = \prod_i \mathcal{N}(0, I)$ as the prior. β reweighs the KL-divergence, which aims to learn the disentangled factors in Z^s .

Graph Contrastive Learning: The original graph G and its augmented positive graph G^+ can be utilized to provide the inter-graph contrastive learning(inter-GCL). We use the GCN network as the encoder $f(\cdot)$. More formally, we have:

$$H = \text{GCN}_\theta(G); H_+ = \text{GCN}_\theta(G^+). \quad (28)$$

Note that G^+ are generated via sampling from the probability $f_G(\cdot, \cdot)$. The sampling process and the normalizing process $\hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}}$ in Eq. 28 take lots of computation. Denote the estimated probability of the augmented matrix as $\hat{A}^s$ learned by $f_G(\cdot, \cdot)$. We simplify the

GCN encoder by substituting the normalizing process by estimating probability directly:

$$H_+^{(l+1)} = \sigma \left((\tilde{A}^+ + I_N) H_+^{(l)} W^{(l)} \right). \quad (29)$$

Only the encoders for augmented positive graphs use the above simplification, which is different from the encoder for graph G . The parameters of encoders for all these graphs are still the same. With the encoder to capture the hidden information, we construct the graph contrastive learning via an MSE loss as follows:

$$\mathcal{L}_A = \frac{1}{N} \sum_{i=1}^N [\|h_i - h_i^+\|_2^2 - \|h_i - h_{j_i}^+\|_2^2], \quad (30)$$

where ξ is the threshold that controls the effective difference, γ balances the importance from the positive view and j_i is different with i and sampled randomly. We measure the mutual information between graph G and augmented graph G^+ simply by their difference, i.e., $I(h, h^s) = \|h - h^+\|_2^2$.

Besides inter-graph contrastive learning, we also introduce intra-graph contrastive learning(intra-GCL). For each node in the graph G , we treat the other nodes with the same auxiliary variable as the positive nodes and negative nodes otherwise. More formally, we have the intra-graph contrastive learning loss as:

$$\mathcal{L}_B = \frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(h_i, \tilde{h}_i))}{\sum_{k=1, s_i \neq s_k}^N \exp(\text{sim}(h_i, h_k))}, \quad (31)$$

where $\tilde{h}_i$ is a positive sample of node i with the same auxiliary variable. Then the RAGA becomes:

$$\min_{\psi} \max_{\theta} \lambda_1 \mathcal{L}_A(\theta, \psi) + \lambda_2 \mathcal{L}_B(\theta) + \mathcal{L}_V(\psi). \quad (32)$$

B Clustering Algorithm

In this section, as shown in Alg. 1, we provide a detailed description of our implementation of the clustering algorithm in PFGE-C. The primary goal of the clustering algorithm we implemented is to serve a verification role. To achieve this, we aim to minimize the complexity of the clustering process as much as possible.

Existing clustering methods typically calculate the similarity of all parameters each time for clustering, which, while effective, comes with relatively high computational overhead. In contrast, our implementation first calculates the centers of all models, averages them to obtain a one-dimensional center list, and then applies the same operation to all client models to produce a one-dimensional client list. We then use the KMeans algorithm for clustering, setting a maximum of 20 iterations. The number of clustering centers is heuristically chosen: 2, 4, and 12 for client counts of 5, 10, and 30, respectively. Because our clustering algorithm is intentionally simplified to prioritize operational efficiency, its experimental performance may not be exceptional. Future work could explore more sophisticated clustering algorithms that better balance performance and computational overhead.

Algorithm 1 Clustering Algorithm

Initialization: max iter t , clusters number n , client number N
Input: client model parameter set ψ
Output: cluster label set L
 // Computational model center
 $\tilde{\psi} = \frac{1}{N} \sum_{i=1}^N \psi_i$
 // Calculate the distance between the client model and the center model
 ψ_i in ψ distance_i = $\cos(\psi_i, \tilde{\psi})$
 // Initialize KMeans parameters.
 KMeans($n_clusters=n$, $max_iter=t$)
 // KMeans is used for clustering distances.
 $L = \text{KMeans.fit}(\text{distance})$
 return L

Table 4: Overall comparisons over three datasets

Algorithm	Cora	Citeseer	Pubmed
Local	68.1±1.14	69.0±1.87	83.5±0.27
FedAVG	80.2±1.17	77.2±0.44	84.1±0.22
FedProx	80.5±1.78	78.1±0.84	85.1±0.16
FedPUB	79.3±0.89	78.3±1.22	85.5±0.60
pFedHN	80.1±0.56	78.1±0.92	83.5±0.09
PFGA-A	83.1±1.34	79.5±1.17	86.1±0.95

C Experimental

C.1 Extremely Imbalanced Division

Experimental Settings. In this experiment, each category is divided into two parts, with each part representing a certain proportion. For example, one part may account for 70% of the data, and the other for 30%. The two parts of each category are then combined, with the corresponding parts from different categories grouped together. The second part is randomly shuffled. Each client is assigned a portion from the first part of the combination (representing the continuous class) and a corresponding portion from the second part (representing the random class). This process results in label-imbalanced data for each client.

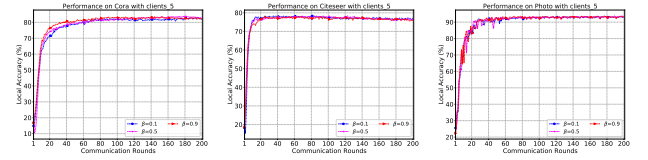
Performance. In node classification tasks with extreme label imbalance, the assignment of nodes to clients often combines a dominant label category with a minor one, resulting in imbalanced data. Local Model Performance: In scenarios with label imbalance, local models generally underperform, as they struggle to handle this imbalance, which degrades their overall performance. Traditional Federated Learning: Conventional federated learning methods (e.g., FedAVG, FedProx, FedPUB, pFedHN) outperform local models in imbalanced tasks, demonstrating their effectiveness in addressing label imbalances. BGA Performance: BGA outperforms traditional federated learning methods in label-imbalanced node classification, showing an enhanced ability to process imbalanced data, as evidenced by significant improvements across various datasets.

Overall, the results indicate that in label-imbalanced node classification tasks, both traditional federated learning methods and BGA outperform local models. These approaches more effectively

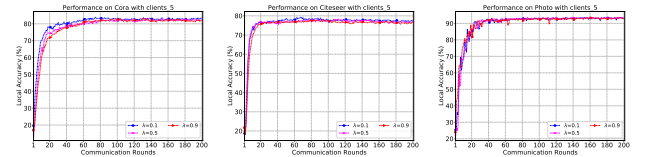
extract useful information from imbalanced data distributions, providing valuable insights for addressing class imbalance challenges in practical applications.

C.2 Parameter Analysis

First, we conduct a sensitivity analysis of the parameter β in Eq.13. The scenario selected for this analysis involves five clients with random data division. The datasets used are Cora, CiteSeer, and Amazon-Photo. We set β to 0.1, 0.5, and 0.9 to observe the model's convergence speed under different values. As shown in Figure 4, when β is close to 0.5, the model is more stable. In contrast, when β is closer to 0.1 or 0.9, the fluctuations are larger, especially in the Amazon-Photo dataset. This behavior can be attributed to an imbalance between the local and global models, which leads to increased fluctuations.

**Figure 4: Parameter sensitivity analysis for β**

Secondly, we conduct a sensitivity analysis of the parameter λ in Eq. 12. The scenario selected for this analysis is the same five-client setup with random data division. The datasets used are Cora, CiteSeer, and Amazon-Photo. In this case, we fix β at 0.5 (as we did in the sensitivity analysis for β , where λ was fixed at 0.5) and vary λ to 0.1, 0.5, and 0.9 to observe the model's convergence speed. As shown in Figure 5, we observe that in most cases, a λ value of 0.1 leads to better results. When λ is set to 0.9, the model converges more quickly. However, in the Amazon-Photo dataset, where accuracy is already high, the differences are less noticeable. Nonetheless, we find that the fluctuations become more pronounced when λ is set to 0.9.

**Figure 5: Parameter sensitivity analysis for λ**

C.3 Baseline Implementation Details

We use a two-layer GCN as the base GNN for all models, with the hidden layer dimension set to 64. Over 200 rounds of federated learning communication were conducted across all datasets. We measure node classification accuracy on the client subgraphs and then compute the average performance across all clients. Given the different design approaches of the models, the number of parameters varies. Therefore, in addition to the basic GNN, we add appropriate network layers to the different models to ensure that the parameter counts are approximately consistent.